

CHARLES OBIKA

Senior Full Stack Engineer | AI & Distributed Systems
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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 9+ years of experience designing, building, and scaling high-throughput distributed systems, AI-driven platforms, and event-driven microservices across FinTech, HealthTech, and Enterprise SaaS. Proven track record delivering mission-critical financial infrastructure — including NIBSS integrations, real-time KYC/AML platforms, and LLM-powered document intelligence pipelines — at production scale. Deep expertise across the full stack: Node.js, Python, .NET Core, React.js, and TypeScript on cloud-native infrastructure (AWS/Azure/GCP), containerised with Kubernetes and Docker. Consistently bridges engineering excellence with business outcomes — achieving 99.99% uptime, 35% API latency reduction, and measurable delivery acceleration through AI-assisted development (Cursor, Claude Code). Seeks a senior or lead engineering role to architect resilient systems, drive engineering culture, and deliver measurable impact.

CORE COMPETENCIES

Full Stack Engineering (Node.js · Python · .NET Core · React.js · TypeScript)
Distributed Systems & Microservices · Event-Driven Architecture · Clean Architecture / DDD
LLM Integration · RAG Pipelines · AI-Assisted Development (Cursor · Claude Code)
High-Throughput API Design (REST · OpenAPI · Webhooks · GraphQL)
Cloud Infrastructure (AWS · Azure) · Containerisation (Docker · Kubernetes · ECS/AKS)
Database Architecture (PostgreSQL · SQL Server · MongoDB · Redis · Vector DBs)
CI/CD Automation · Observability (OpenTelemetry · Serilog · Dynatrace)
Security & Compliance (PCI DSS · OWASP · JWT · OAuth2 · KYC/AML)
Technical Mentorship · Cross-Functional Leadership · Agile (Scrum/Kanban)

TECHNICAL SKILLS

Category	Technologies
Languages	Python, C#, JavaScript, TypeScript, Node.js, SQL, VB.NET
Frontend	React.js, Next.js, HTML5, CSS3, Tailwind CSS, Redux
Backend & Frameworks	ASP.NET Core, FastAPI, Flask, Django, Express.js, NestJS
AI & LLMs	OpenAI / Azure OpenAI, LangChain, RAG, Embeddings, Vector DBs (Pinecone, Qdrant), Cursor, Claude Code
Databases	PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Vector DBs
Cloud & Containers	AWS (Lambda, ECS, S3, IAM, RDS), Azure (App Services, Functions, AKS, OpenAI), Docker, Kubernetes, Terraform
DevOps & CI/CD	GitHub Actions, Jenkins, Azure DevOps, Nginx, Dynatrace, OpenTelemetry, Serilog
Messaging & Streaming	RabbitMQ, Kafka, Celery, Async Workers, Background Tasks
APIs & Integration	REST, OpenAPI/Swagger, Webhooks, OAuth2, Payment Gateways (NIBSS, Stripe, PayPal, Paystack)

Category	Technologies
Security & Compliance	PCI DSS, OWASP Top 10, JWT, OAuth2, Rate Limiting, KYC/AML, Data Encryption
Tools & Platforms	GitHub, Bitbucket, JIRA, Confluence, Postman, Agile (Scrum/Kanban)

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Full Stack & AI

Premium Trust Bank · Nigeria · *March 2025 – Present*

Lead full stack and backend engineer for high-throughput financial transaction, identity, and customer-facing systems processing millions of daily operations. Architecting and optimising .NET Core and Python microservices under strict PCI DSS and CBN regulatory compliance while contributing to frontend engineering on internal banking portals.

- Reduced API p95 latency by 35% through Redis caching, async processing, and query optimisation across PostgreSQL and SQL Server — directly improving mobile app responsiveness for 200,000+ customers
- Designed and deployed multi-region microservices on Azure AKS, supporting distributed financial workloads with automated failover, circuit breakers, and 99.99% uptime SLA
- Built real-time KYC/AML integration layer connecting core banking to multiple verification providers with webhook processing, retry logic, and caching — reducing customer onboarding time by 45%
- Engineered secure REST APIs with OAuth2, JWT, fine-grained rate limiting, and comprehensive audit logging — meeting PCI DSS and CBN compliance requirements
- Implemented NIBSS NRS Webhook Notification Handler for real-time transfer alerts, enabling automated reconciliation and instant customer notifications
- Led .NET 6 → .NET 10 platform migration across 10+ microservices and NuGet packages — resolving breaking changes, dependency conflicts, and CI/CD pipeline failures across Azure DevOps
- Accelerated team delivery by 30% through AI-assisted development (Cursor + Claude Code) for debugging, refactoring, code generation, and automated test authoring
- Mentored junior engineers through design reviews, ADRs, pair programming, and structured code review processes

Stack: .NET Core 10, Python, React.js, TypeScript, PostgreSQL, SQL Server, Redis, Azure AKS, Azure DevOps, RabbitMQ, JWT, OAuth2, OpenTelemetry, Serilog

Senior Backend Engineer — AI & Distributed Systems

Turing · Remote (USA) · *February 2025 – August 2025*

Delivered AI-powered microservices and production LLM pipelines for intelligent document processing, structured reasoning, and semantic search systems — collaborating across US/EU time zones in a fully remote, Agile engineering organisation.

- Architected production RAG pipeline using FastAPI, OpenAI embeddings, Pinecone/Qdrant vector databases, and Redis caching — processing 10,000+ documents/month with 94% structured data extraction accuracy
- Reduced LLM inference response time by 74% (4.2s → 1.1s) by engineering scalable async workers (FastAPI + Celery + Redis) for high-throughput inference queuing

- Reduced document retrieval latency by 62% through semantic chunking, embedding optimisation, and vector index tuning on Pinecone
- Authored OpenAPI specifications, PRDs, and system design documents for distributed AI services consumed by internal and external engineering teams
- Deployed production services via GitHub Actions CI/CD, Docker, and Kubernetes with zero-downtime rolling deployments

Stack: Python, FastAPI, OpenAI API, LangChain, Pinecone, Qdrant, Redis, Celery, Docker, Kubernetes, GitHub Actions

Senior Full Stack Engineer

EgolePay · Nigeria (FinTech) · November 2021 – February 2025

Architected and delivered high-throughput financial transaction microservices for a payment platform processing millions of daily requests. Led migration from monolithic architecture to event-driven microservices and contributed full-stack across backend APIs and merchant-facing web portals.

- Designed event-driven transaction orchestration engine using RabbitMQ and .NET Core — decoupling payment processing, reconciliation, and notification services and improving fault tolerance by 40%
- Handled 500+ TPS peak throughput with idempotent APIs, compensating transactions, dead letter queues, and automated daily reconciliation pipelines
- Integrated NIBSS EasyPay and multiple payment gateways — engineering secure, idempotent webhook handlers and automated settlement workflows reducing settlement time from 24 hours to near real-time
- Built developer-friendly REST APIs consumed by mobile apps, merchants, and third-party partners — with full OpenAPI documentation, semantic versioning, and self-service onboarding flows
- Led monolith-to-microservices migration to Kubernetes (EKS) with API gateway, service mesh, and distributed tracing — reducing release lead time from 2 weeks to 3 days
- Reduced infrastructure costs by 25% through Kubernetes HPA right-sizing, EKS workload optimisation, and RDS query tuning
- Achieved 99.99% availability with circuit breakers, retry policies, dead letter queues, and full observability stack (OpenTelemetry + Serilog + Dynatrace)
- Built merchant-facing React.js portals for transaction monitoring, reporting, and onboarding — improving merchant self-service adoption by 35%
- Mentored 3 engineers, increasing team velocity by 20% through structured code reviews, knowledge sharing sessions, and architectural decision records

Stack: .NET Core, Node.js, React.js, TypeScript, PostgreSQL, RabbitMQ, Redis, Kubernetes (EKS), AWS (S3, IAM, RDS), OpenTelemetry, Serilog, Dynatrace, Docker, GitHub Actions

Senior Software Engineer — Full Stack

Archimedes · Remote (USA) · February 2022 – July 2022

Built scalable backend services, REST APIs, and frontend components for e-commerce and social platforms serving 500,000+ monthly active users across a distributed, multi-team engineering environment.

- Developed secure payment integrations (Stripe, PayPal) with webhook handling, idempotency keys, and automated refund workflows — achieving zero payment processing incidents during tenure
- Engineered scalable media handling infrastructure using presigned URLs (AWS S3) and image optimisation pipelines — reducing media load time by 40%

- Contributed to React.js frontend for product listing, checkout flows, and user dashboard features
- Maintained OpenAPI documentation and contributed to semantic versioning and release workflows across multiple engineering teams

Stack: Node.js, React.js, TypeScript, PostgreSQL, AWS S3, Stripe, PayPal, Docker, OpenAPI

Software Engineer — Full Stack

Courteville Business Solutions · Nigeria · *June 2021 – November 2021*

Developed automated payment, billing, and valuation modules across multiple enterprise web applications — contributing across backend APIs and frontend interfaces.

- Built REST APIs for utility payments, third-party data integrations, and subscription lifecycle management serving enterprise clients
- Refactored legacy VB.NET modules to .NET Core — improving test coverage by 25% and reducing production defects
- Contributed to React.js frontend features for enterprise billing and reporting dashboards

Stack: .NET Core, Node.js, React.js, SQL Server, REST APIs, VB.NET

Software Engineer

SYCOM EBS Technology · Nigeria · *January 2020 – June 2021*

Engineered backend APIs and full-stack features for payment systems and enterprise health applications — supporting digital transformation initiatives across healthcare and utilities.

- Refactored legacy monolithic systems into modular, maintainable microservices — improving system performance by 30% and reducing time-to-deploy
- Integrated third-party SMS and email notification services for patient reminders, appointment scheduling, and billing alerts
- Built healthcare data management interfaces for patient records, appointments, and clinical workflows

Stack: .NET Core, Node.js, SQL Server, REST APIs, MongoDB, SMS/Email Integrations

Software Engineer

ITANDT Solutions · Nigeria · *September 2018 – January 2020*

Built reusable backend components and full-stack integration modules for healthcare SaaS applications — contributing to both API development and frontend feature delivery.

- Migrated legacy systems to .NET Core and SQL Server — improving system reliability, modularity, and long-term maintainability
- Developed patient-facing web interfaces for appointment booking, health records access, and billing management
- Built reusable API integration modules consumed across multiple SaaS product lines

Stack: .NET Core, JavaScript, SQL Server, REST APIs, HTML5, CSS3

Software Engineer

Fidelity Bank · Nigeria · *February 2017 – March 2018*

Contributed to .NET application development, backend feature enhancements, and QA processes for enterprise banking systems — gaining foundational experience in financial software engineering and regulatory compliance.

- Supported backend development on internal banking tools including transaction

- reporting, account management, and customer-facing features
- Contributed to API documentation, QA automation, and performance improvements for internal and customer-facing banking applications

Stack: .NET, SQL Server, VB.NET, REST APIs

KEY PROJECTS

LLM-Powered Document Intelligence Pipeline · *Turing* Designed and deployed production RAG system using FastAPI, OpenAI, Pinecone, and Redis. Achieved 94% structured data extraction accuracy across 10,000+ documents/month with 62% latency reduction.

Event-Driven Payment Orchestration Engine · *EgolePay* Architected microservices transaction engine with RabbitMQ, .NET Core, and PostgreSQL handling 500+ TPS peak. Reduced settlement time from 24 hours to near real-time with automated reconciliation.

Monolith to Microservices Migration · *EgolePay* Led migration of legacy payment gateway to Kubernetes (EKS) with API gateway, service mesh, and distributed tracing. Reduced release lead time from 2 weeks to 3 days.

Real-Time KYC Integration Platform · *Premium Trust Bank* Built high-availability integration layer connecting core banking to multiple KYC/AML providers with circuit breakers, caching, and real-time webhooks. Reduced customer onboarding time by 45%.

Premium Mobile — Digital Banking Application · *Premium Trust Bank* Full-stack engineer on Nigeria's Premium Trust Bank mobile banking platform — delivering virtual card lifecycle management, investment booking workflows (Treasury Bills, FGN Bonds), maker-checker approval systems, and admin notification pipelines.

Blayz Financial Ecosystem · *Blayz* Lead backend engineer on multi-service financial ecosystem spanning wallet management, savings, investments, marketplace, ride-hailing, and planner — built on .NET 10 microservices architecture with MongoDB, SQL Server, and Azure.

CERTIFICATIONS

Certification	Year
IBM RAG and Agentic AI Professional Certificate	2026
Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI	2026
AWS Certified Solutions Architect – Associate	2024
Certified PCI Security Manager	2023
IBM Certified Full Stack Developer	2022
Jobberman Soft Skills Program	2021

EDUCATION

Bachelor of Science (B.Sc.) — Computer Science & Information Technology Bowen University, Nigeria · 2018

ADDITIONAL

- Languages: English — Full Professional Proficiency (business, technical, and stakeholder communication)